



Analyzing Market Dynamics & Revitalizing Brand Strategy for Injectable Anesthesia Products

BIA 810
Health Care Data & Analytics
Final Project

Neural Numb-ers

Spring 2026

Today's objectives & agenda



Time



Topic



Presenters

[0:00 – 5:00]

Introduction & Team Introductions

Neural Numb-ers

[5:00 – 7:30]

Where are we Now?

Jeel

[7:30 – 21:30]

3-Staged Solution Recommendation

**Sherwin, Frederick,
Preetham**

[21:30 – 25:00]

Conclusion– finalized recommendations.

Poojan



Meet the Team!



Professor: Sanjiv Koshal



Frederick Blake



Jeel Patel



Sherwin Christian



Preetham Deepak
Kumar



Poojan Mehta



Executive Summary

PROBLEM STATEMENT:

Product 1, the market-leading injectable anesthesia drug, steadily lost market share from 2016–2018, while Product 2 failed to capture the expected cannibalization opportunity. Meanwhile, competitor Product 3 rapidly gained claims volume, patient share, and HCP adoption across key territories.

BUSINESS SOLUTION

We propose a 3-phase strategy focused on:

- **Retaining existing Product 2 writers**
- **Reclaiming shared Product 2/Product 3 writers**
- **Expanding Product 2 adoption among high-value specialties**

BUSINESS IMPACT

This strategy will:

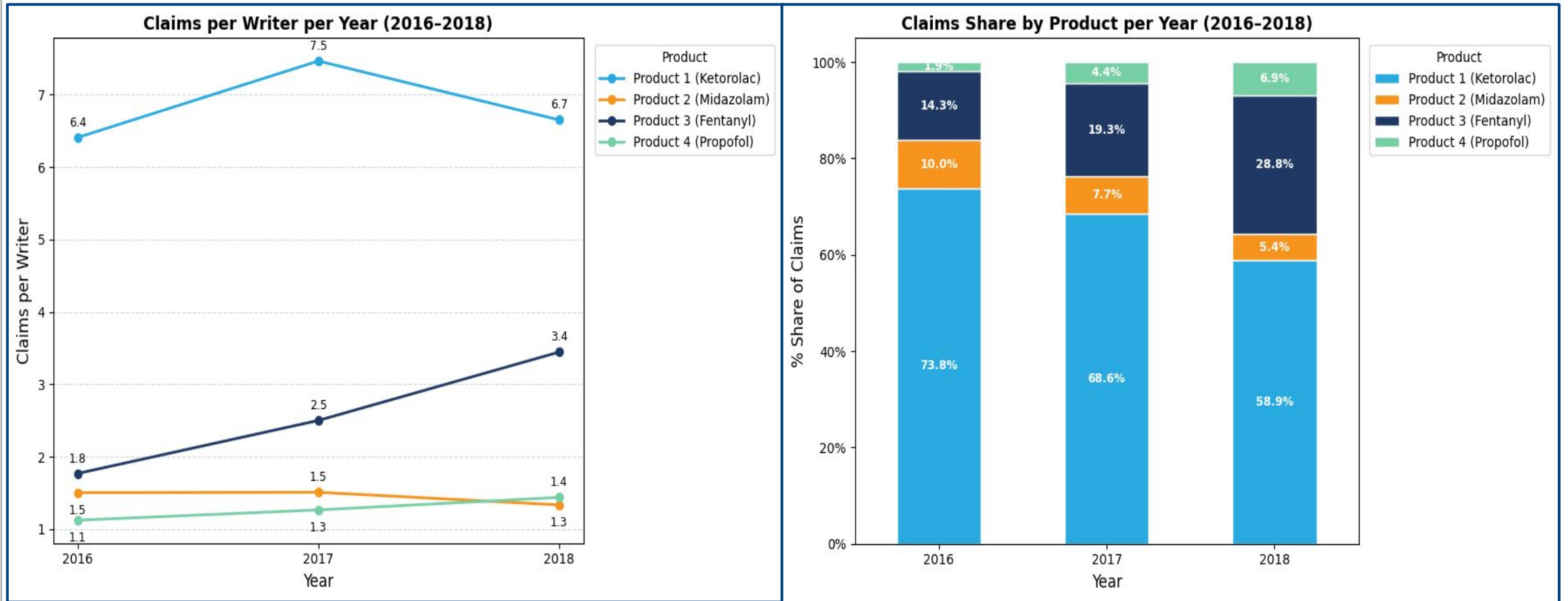
- **Recover lost Product 2 claims volume**
- **Improve HCP retention and engagement**
- **Increase sales force efficiency in high-risk territories**
- **Slow market share erosion and support long-term growth**



Where are we now?



What the Data Tells Us

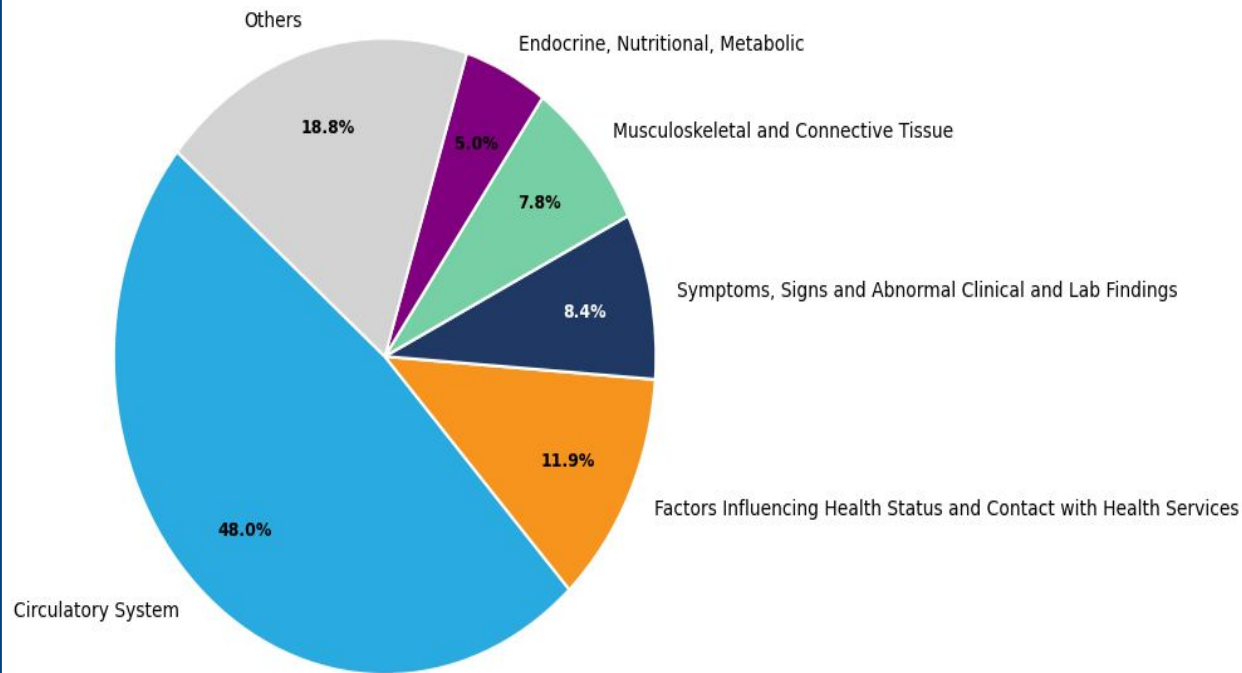


Key Observation: Product 3's claims per writer grew from 1.8 to 3.4 while Product 2 stayed flat (1.5 to 1.4). Product 2's claims share fell from 10.0% to 5.4% as Product 3 nearly doubled its share from 14.3% to 28.8%. Existing writers are not leaving Product 2 -- they are writing it less while writing Product 3 more. This points directly to a writer engagement problem addressed in Phase 1.

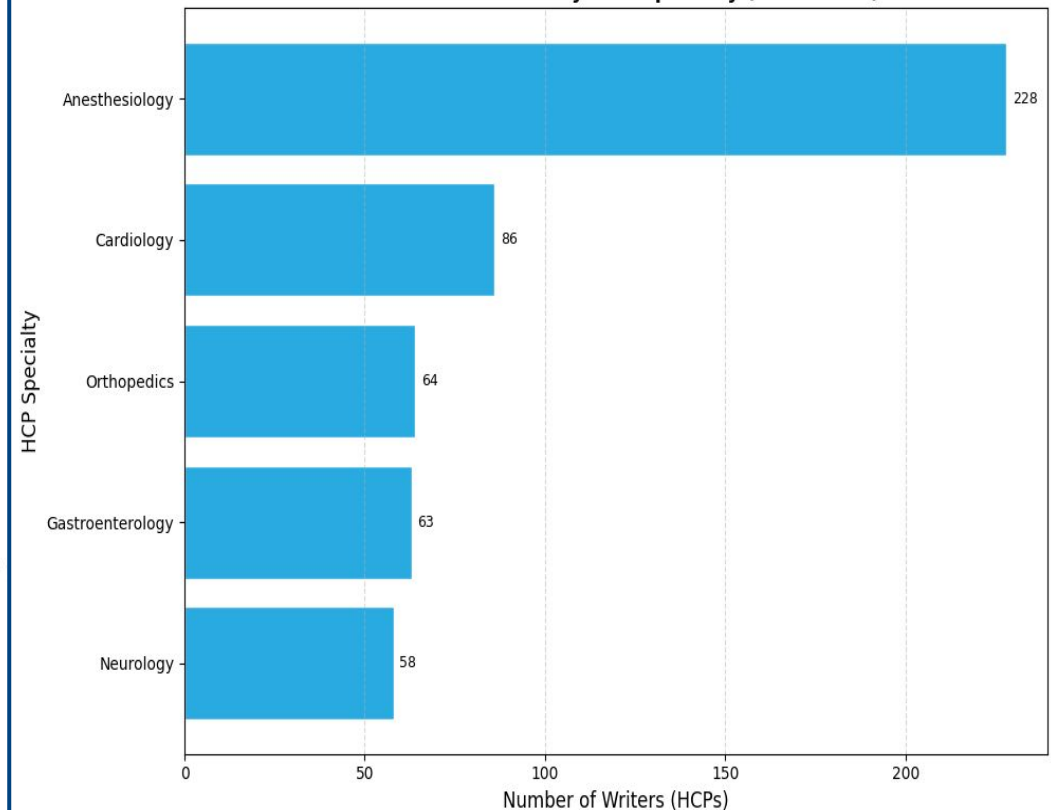


Key Market Drivers

Top 5 Diagnosis Specialties by % of Claims (2016-2018)



Number of Writers by HCP Specialty (2016-2018)



Key Observation: Anesthesiology dominates with 228 writers -- the largest HCP pool -- yet Product 2 penetration remains below 50% within this specialty. Circulatory System accounts for 48% of diagnosis claims. Gastroenterology and Neurology are the most clinically aligned specialties for Product 2's indications, pointing directly to the specialist targeting opportunity in Phase 3.



3-Phase Strategy

Phase 1: HCP Retention

Target HCPs currently writing Product 2 Rx to diminish current loss trend.

Phase 2: Reclaim Shared Writers

Target HCPs that write both Product 2 & Product 3, but favor Product 3.

Phase 3: Specialist Targeting

Target primary specialties utilizing related products to increase Product 2's adoption.



Phase 1: HCP Retention



HCP Retention: largest recovery opportunity

Product 2 has established writer relationships, but retention losses are concentrated within several high-value territories.

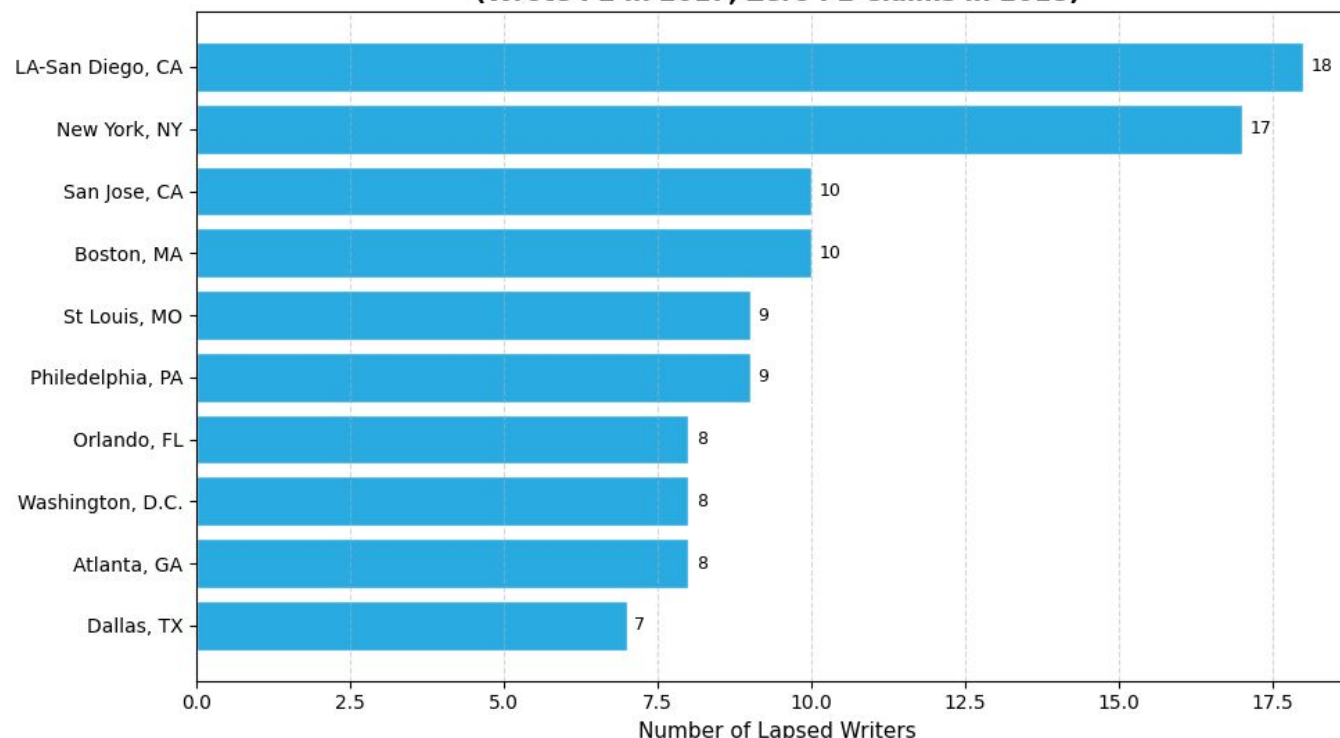
Key readout

84

lapsed Product 2 writers identified between 2017–2018

- LA-San Diego and New York experienced the highest concentration of lapsed Product 2 writers
- Multiple high-volume territories lost significant Product 2 prescribing activity
- Existing sales relationships reduce reactivation barriers and recovery timelines

Top 10 Territories by Lapsed Product 2 Writers
(Wrote P2 in 2017, Zero P2 Claims in 2018)



Recommended action: prioritize reactivation campaigns and field-force engagement within the highest-loss territories before broader expansion efforts



Territory-Level Retention Risk

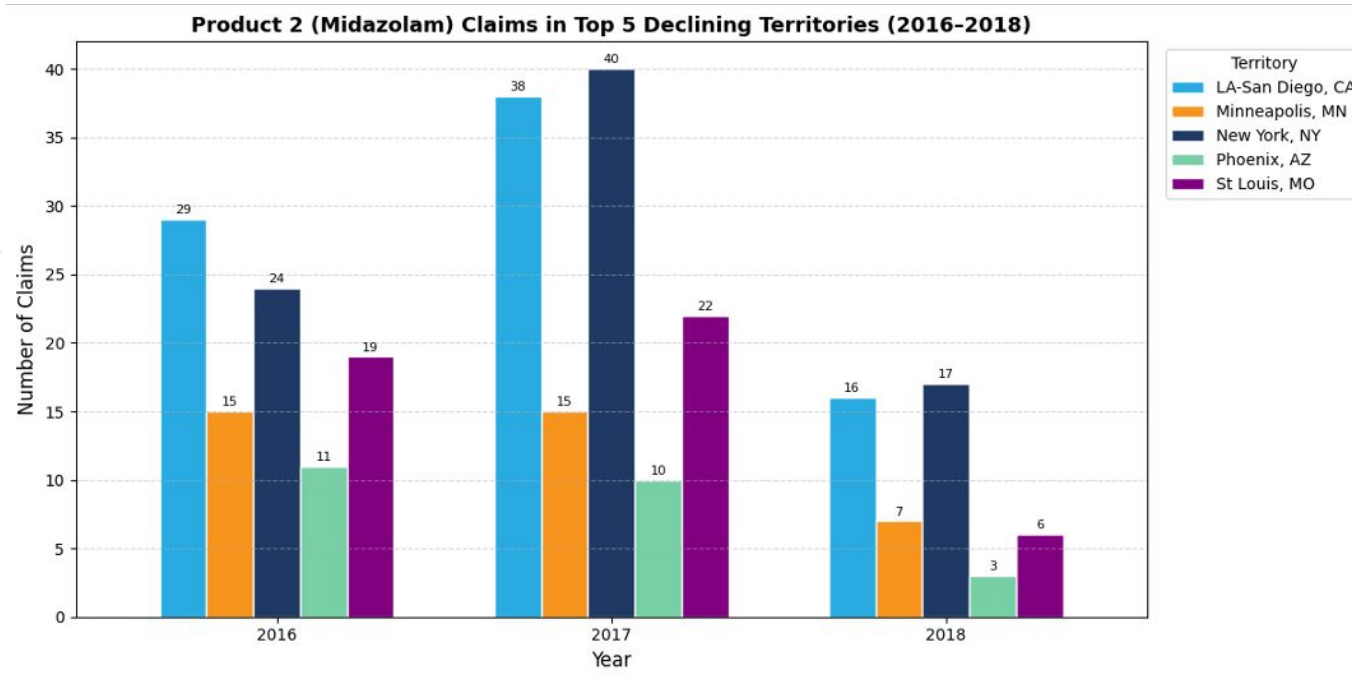
Product 2 experienced its largest declines within several high-volume territories between 2017–2018.

Key readout

5

highest-risk territories show major Product 2 erosion

- Biggest declines occurred in New York, LA-San Diego, Phoenix, St. Louis, and Minneapolis
- Territory-level losses closely mirrored Product 3 growth patterns
- High-volume markets experienced the sharpest retention decline



Recommended action: prioritize field-force reallocation and HCP re-engagement in the highest-risk territories.

Writer Retention vs Competitive Pressure

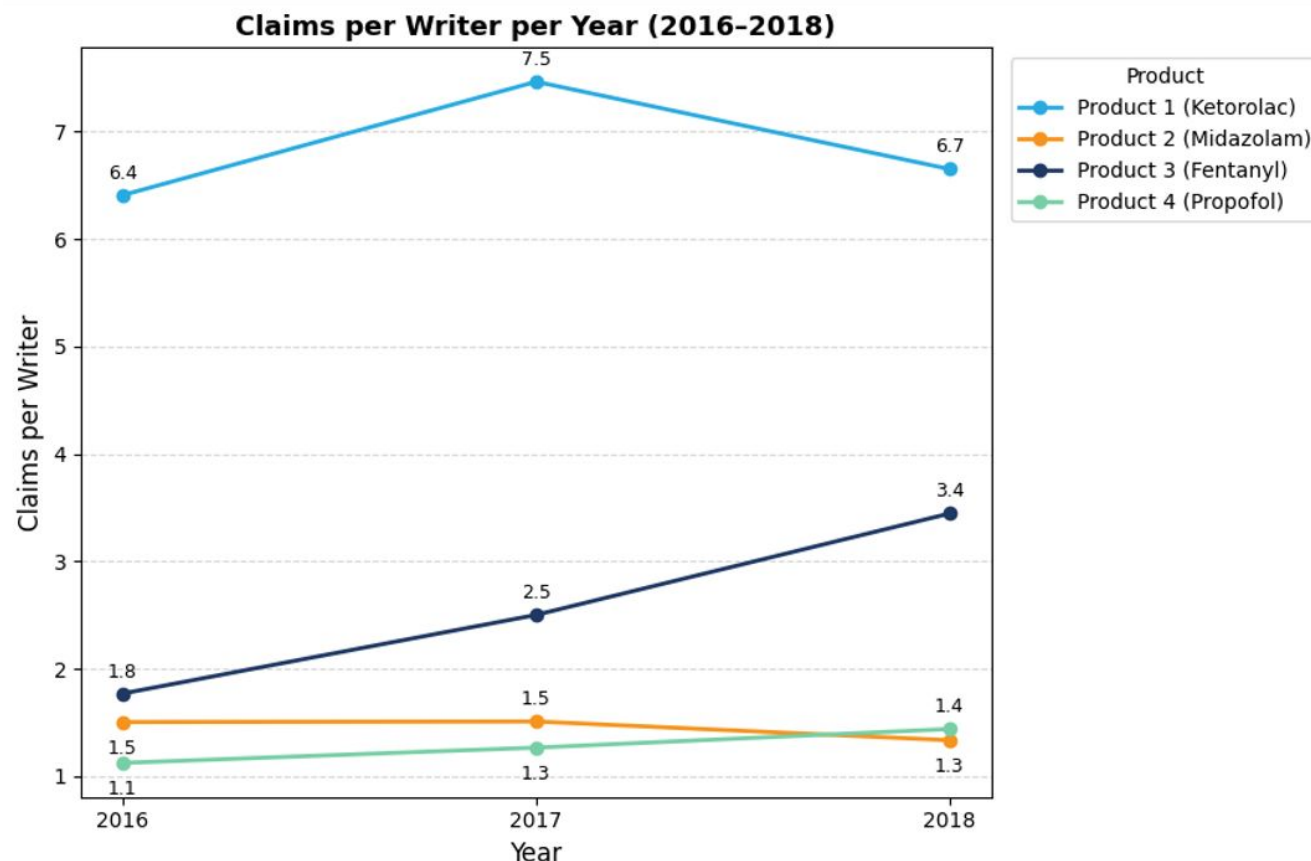
Product 3 continued gaining writer intensity while Product 2 prescribing depth remained weak.

Key readout

20%

potential recovery of lost Product 2 claims volume through targeted writer reactivation

- Product 3 continued gaining claims and writer share between 2016–2018
- Product 2 writer retention weakened despite existing market presence
- Shared HCPs increasingly favored Product 3 prescribing patterns



Recommended action: deploy targeted KOL programs and retention campaigns to slow Product 2 market share erosion.

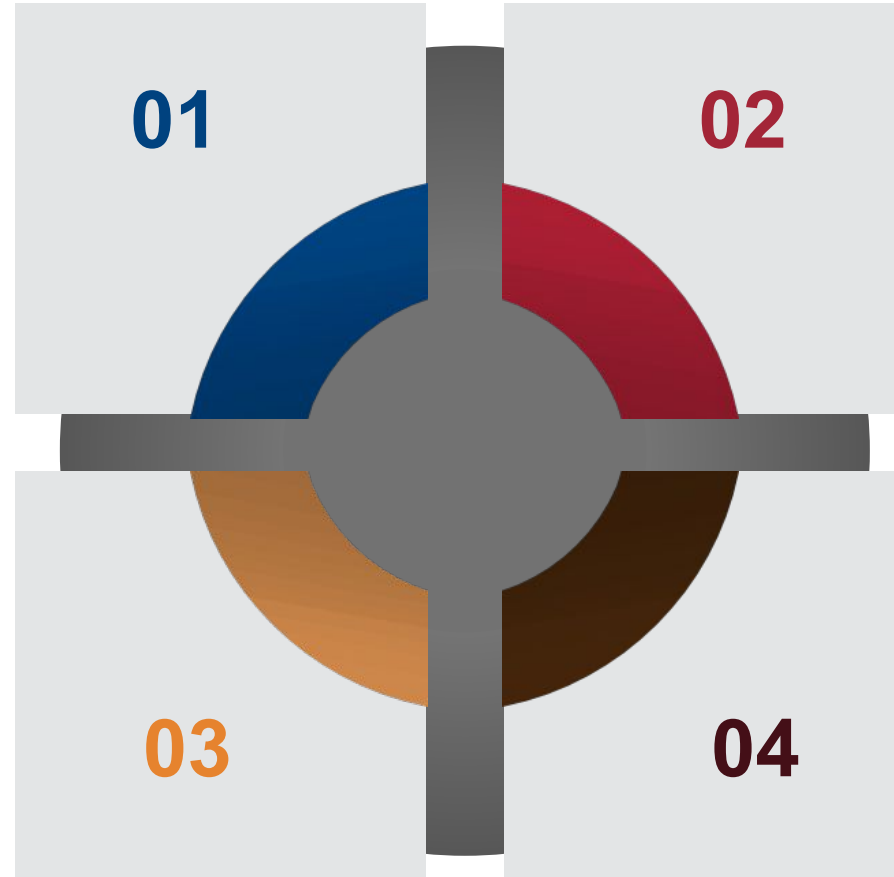
Phase 1: HCP Retention

IMPACT FORECAST

Reactivating 40 of 84 lapsed writers at 2017's average rate recovers ~60 claims— a 20% recovery of the 2017 to 2018 volume loss. Stops the continuing writer decline from accelerating further into 2019.

COST TO DELIVER

Field force reallocation requires no new hires. Two KOL speaker events in New York and LA-San Diego estimated at \$30-50K. Total estimated investment: \$40-70K.



FEASIBILITY

Uses existing reps, existing relationships, and existing claims data. No new materials or indications required. Primary risk: some lapsed writers may be fully committed to Product 3.

TIME TO DELIVER

Lapsed writer list can be pulled from claims data immediately. First reactivation calls deployable within 30 days. Measurable KPI results expected within 90 days.

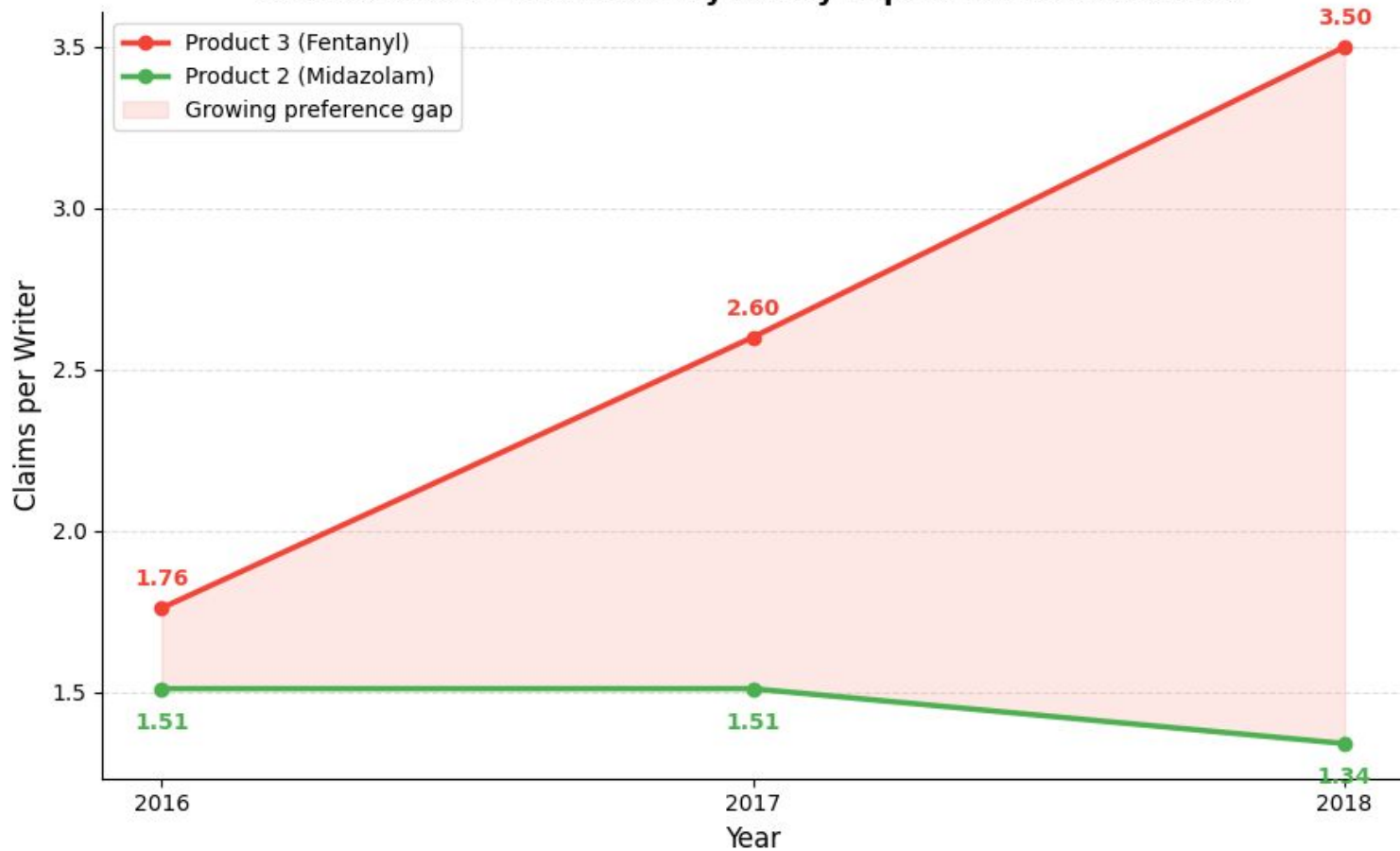


Phase 2: **Reclaim Shared Writers**



Growing Preference Rift - P2 v. P3

Claims per Writer: P2 vs. P3 Among 428 Shared Writers (2016-2018)
Same doctors — P3 intensity nearly tripled while P2 declined



- Every HCP that prescribes P2 also prescribes P3.
- Represented here: Prescribers who prescribe *both* P2 & P3.
- These HCPs are *aware* of P2, but prefer to prescribe P3.

Competitor Comparison

Midazolam Hydrochloride (P2)

“Intravenously injected for induction of general anesthesia.”

Primary Use-Cases:

- Seizures
- **Anesthesia in surgery**
- Anxiety Disorders (Drugbank)

Fentanyl Citrate (P3)

“For analgesic action of short duration during the anesthetic periods, premedication, induction, and maintenance, and in the immediate postoperative period (recovery room) as the need arises.”

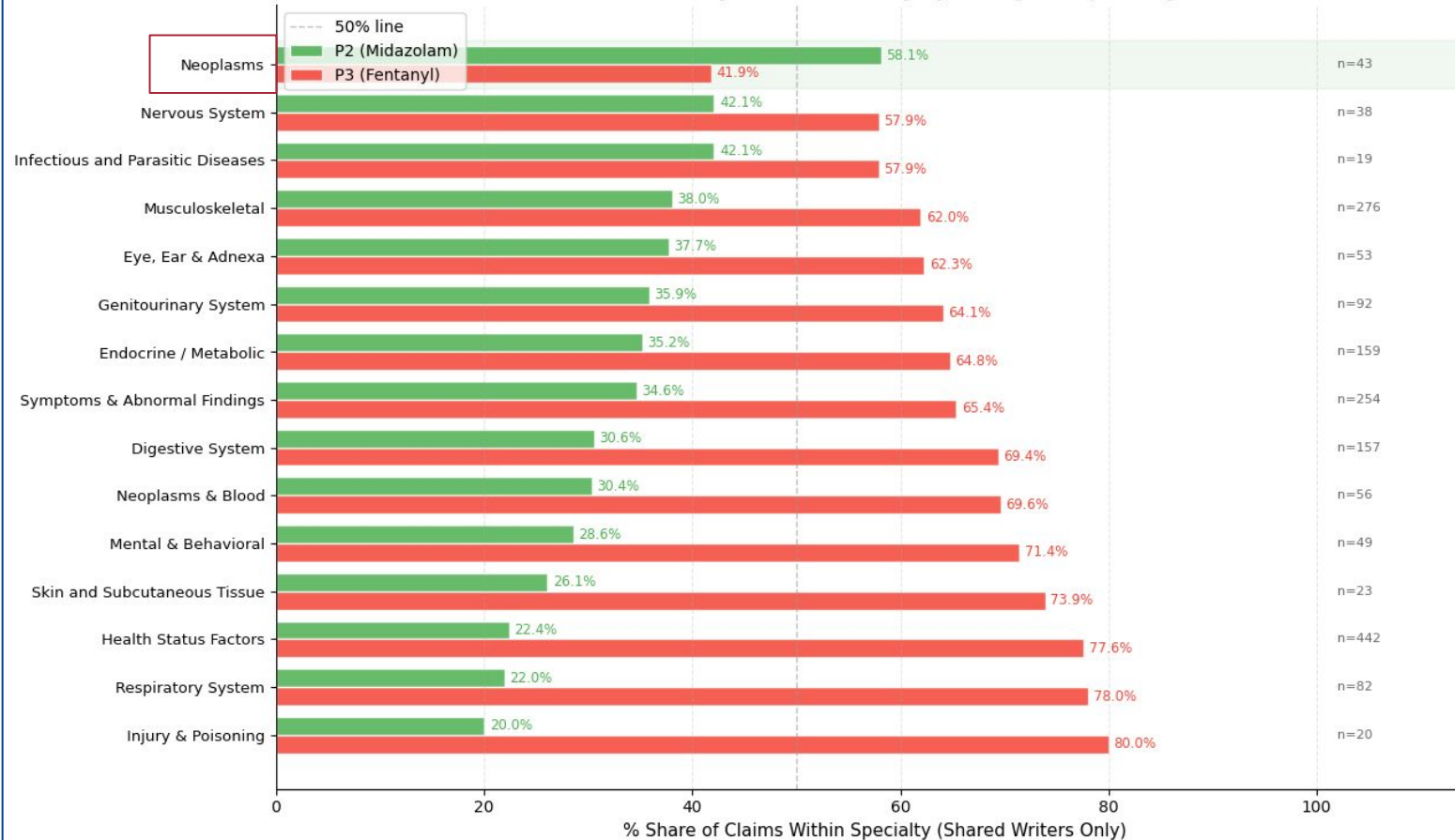
Primary Use-Cases:

- Cancer pain
- **Anesthesia in surgery**
- Chronic pain (Drugbank)



P3 Dominates Nearly Universally

P2 vs. P3 Claim Share by Diagnosis Specialty — Among 428 Shared Writers
P3 dominates every non-circulatory specialty except Neoplasms



58.1% dominance over P3 in Neoplasms.

Neoplasm: An abnormal growth of tissue, either determined to be cancerous or non-cancerous.

Conclusion: P2 *can* win when the clinical case is made correctly. Sales must overcome a clear ***clinical preference shift***.

Target fields that P2 specializes in—seizures, surgery, and anxiety.

Note that in many of these fields, the use of P2 & P3 in synergy is a common approach (Stability).

Specialty Targeting: largest unconverted writer pools

Product 2 has a clinical use difference to Product 3— emphasize it as a strength rather than a weakness.

Key readout

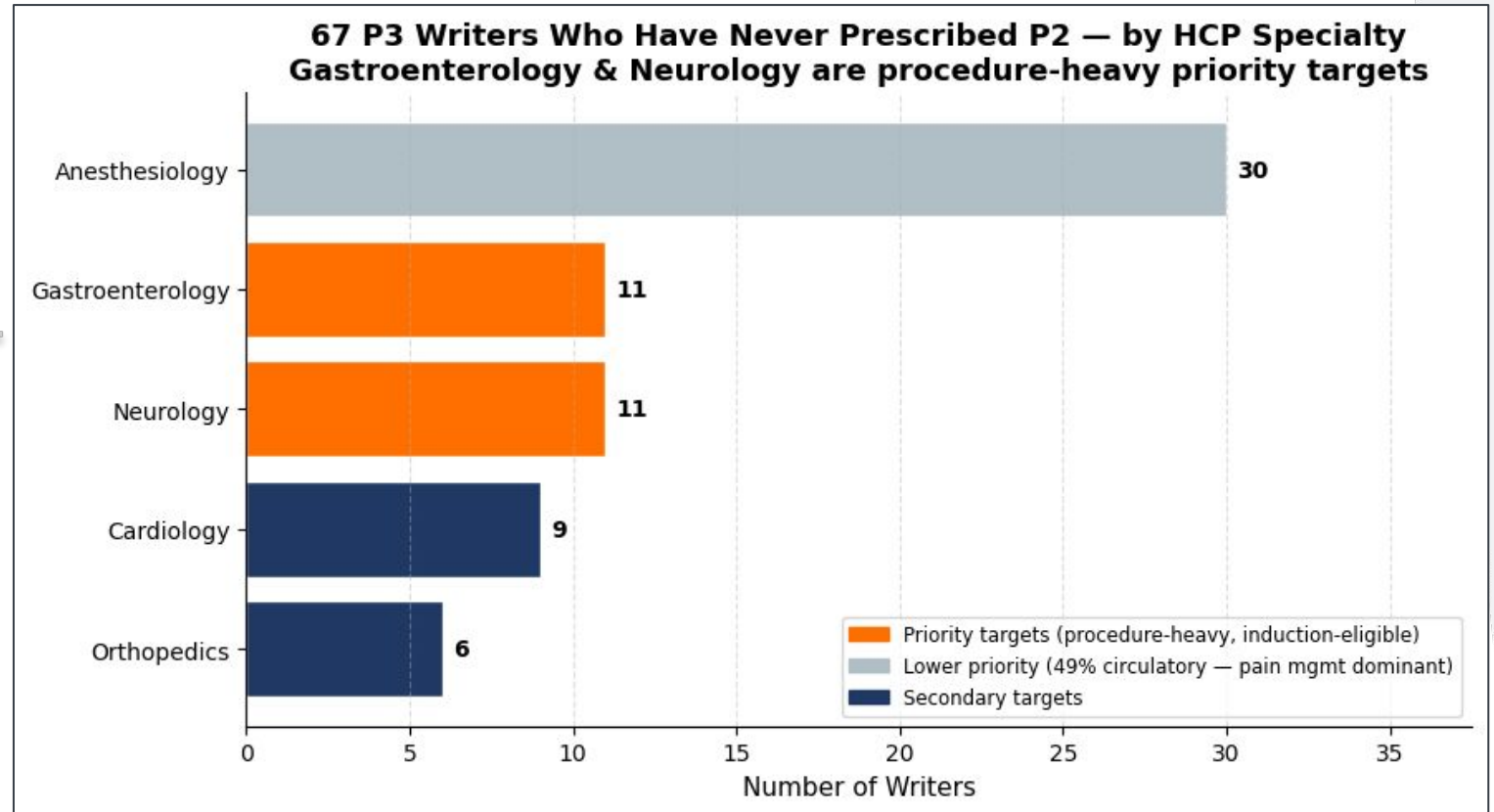
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priority HCP targets within fields of viable applications.

- Gastroenterology and Neurology (low-pain surgery, seizures, and anxiety) are primary use-cases for product 2.
- Anesthesiology has a major HCP pool and— while it leans heavily on P3's pain relieving properties— also represents a sizeable opportunity market.

Opportunity for P2 Adoption:

1. In conjunction with P3 use
2. Independently for low-pain circumstances



Recommended action: Target HCPs in fields with a primary use case for Midazolam (Musculoskeletal and Digestive).

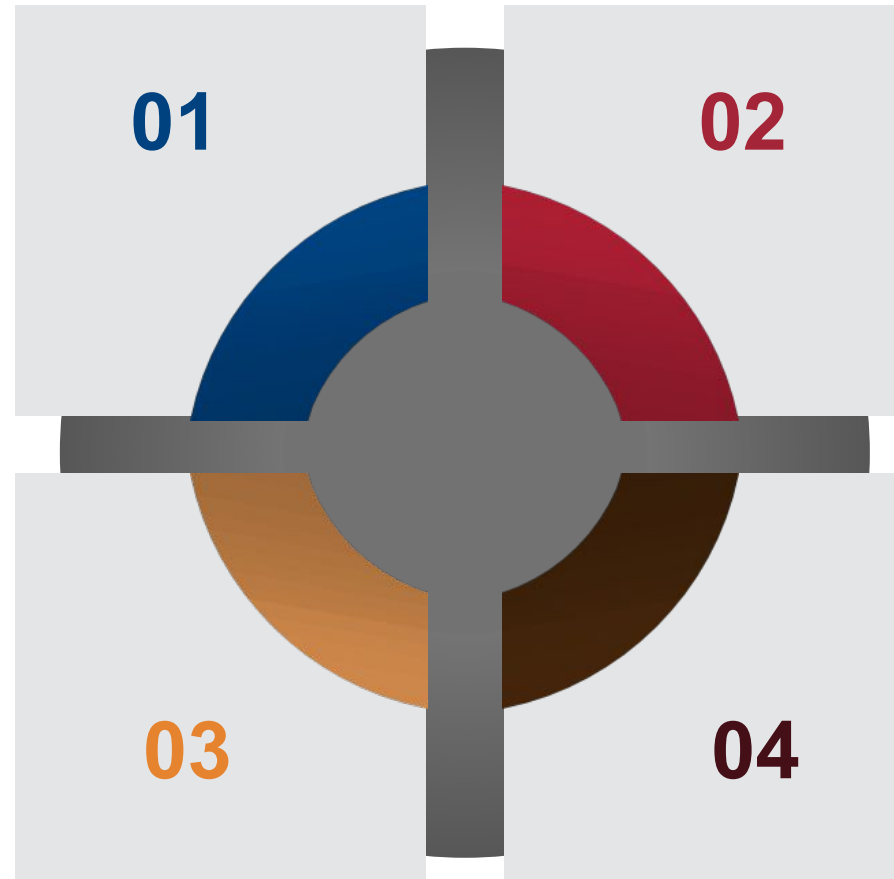
Phase 2: Reclaim Shared Writers

IMPACT FORECAST

Restoring 428 shared writers from 1.34 to the 2016 baseline of 1.51 CPW yields approximately 73 additional claims— a 24% recovery. Each 0.1 CPW gain across the shared writer base equals roughly 43 incremental claims annually.

COST TO DELIVER

Requires development of new indication-specific clinical detail aids contrasting P2 induction vs. P3 analgesia by procedure type. Ongoing rep training required to deliver a more nuanced clinical argument.



FEASIBILITY

The 428 writers are reachable but require a new clinical narrative. Neoplasms' 58.1% P2 share proves the message can work; replication across Musculoskeletal and Digestive is the execution challenge.

TIME TO DELIVER

New clinical materials require 60-90 days to develop and approve through medical/legal review. Full deployment to 428 shared writers realistically 90-120 days.



Phase 3:

Specialist Targeting



Specialty Targeting: largest unconverted writer pools

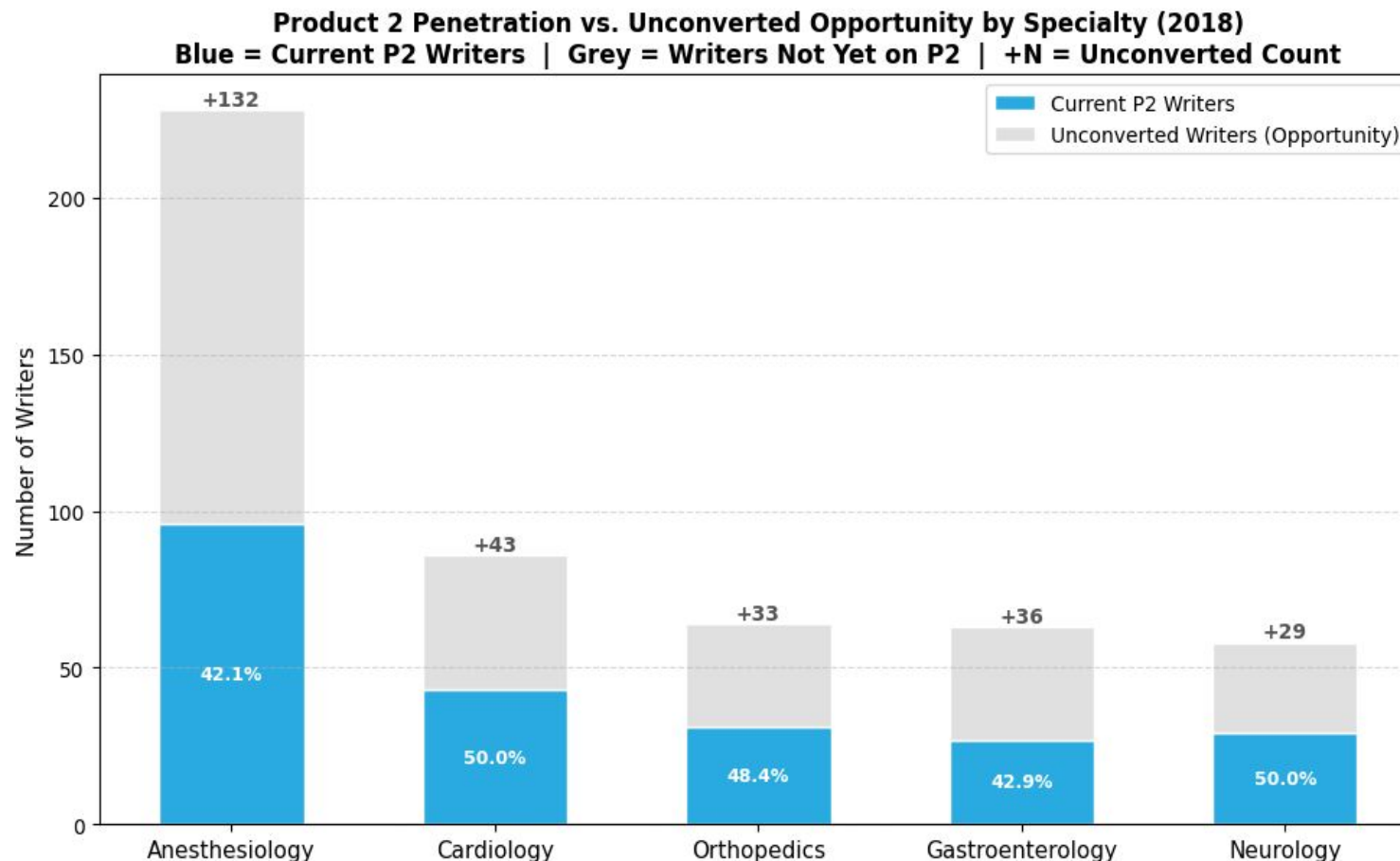
Product 2 has established market presence, but major specialties still represent significant untapped growth opportunity.

Key readout

273

unconverted writers across the five largest target specialties

- Anesthesiology represents the largest opportunity with **132 unconverted writers**
- Product 2 penetration remains below **50%** across key specialties
- Prioritize **Anesthesiology** first, followed by **Gastroenterology** and **Orthopedics** as the second wave



Recommended action: convert 35 specialists in the first wave to create a measurable growth bridge after retention stabilizes.

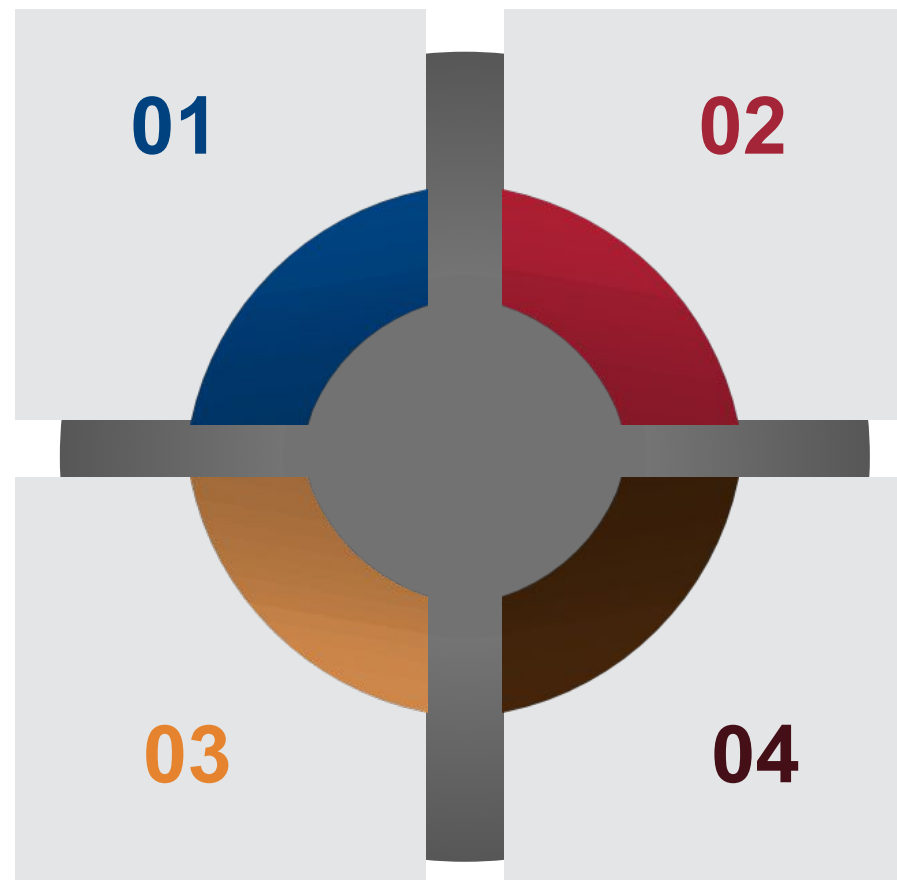
Phase 3: Specialist Targeting

IMPACT FORECAST

Add 20 Anesthesiology writers & 15 Gastro/Ortho writers generates ~52 new annual claims. Reduce dependence on territories / writers being contested by P3.

COST TO DELIVER

Two specialty-specific detail aids, digital NPP targeting, and rep training. Estimated ~\$100-200K cost.



FEASIBILITY

Contingent on Phase 1 stabilizing currently dwindling writer base. Clinical fit is strong, but requires larger chronological and monetary investment.

TIME TO DELIVER

Planning begins, execution after Phases 1 & 2's measurable results. New specialty writers expected at 90-day mark, full results after 12-month.

3-Phase Strategy

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Phase 3: Specialist Targeting

Target primary specialties utilizing related products to increase Product 2's adoption.



Q&A





Thank you!

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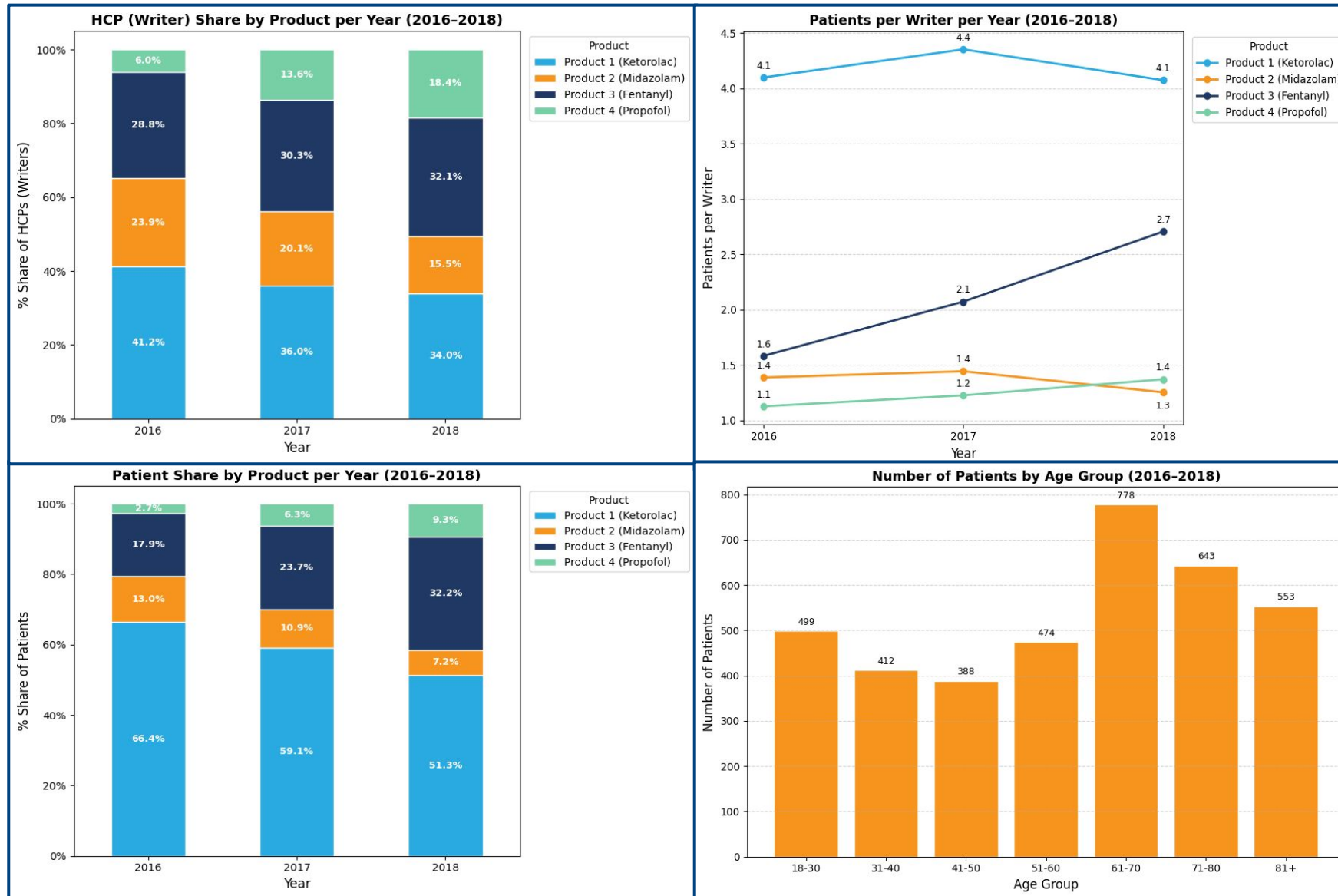
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Unused Data Visualizations (1/2)



Unused Data Visualizations (2/2)

